# Donor health screening, step by step

Before any blood is taken, BloodLink makes sure the donor is safe to give and the blood is safe to receive. A member can answer the health questions online first; at the bank a hospital officer repeats them, takes the vital signs and runs the rapid tests for blood-borne infections. BloodLink applies the deferral rules and will **only record a donation after an eligible screening on the same day**. This guide covers the flow, the rules, what each person sees, and a test script (the same flows as Sections 10 M13, O4, O7, O12 and O13 of the QA Manual).

01 / Who does what

## Member, officer and the system

MEMBER · SCREENING.PHP

### Health Check (optional, 1 minute)

Sidebar → **Health Check**. Choose sex (women also see the pregnancy question) and genotype if known, answer 14 yes/no questions, click **Check my answers**. The answer is immediate: nothing stops you, wait until a date (with the reason), or speak to the bank. Donate Blood shows the status; the check can be retaken.

OFFICER · SCREEN_DONOR.PHP

### Screen Donor (required, at the bank)

Sidebar → **Screen Donor** (by Member ID or email), or **Screen donor** next to a donor on Match Donor. Confirm the pre-filled answers with the donor, enter weight, haemoglobin, blood pressure, pulse and temperature, mark the five rapid tests Negative or Reactive, and click **Save screening**.

SYSTEM · BL_EVALUATE_SCREENING()

### Applies the rules

Refuses incomplete screenings, then decides **eligible**, **deferred until a date**, or **permanently deferred**; when several findings apply the longest deferral wins. Stores everything in `donor_screenings` and notifies the member.

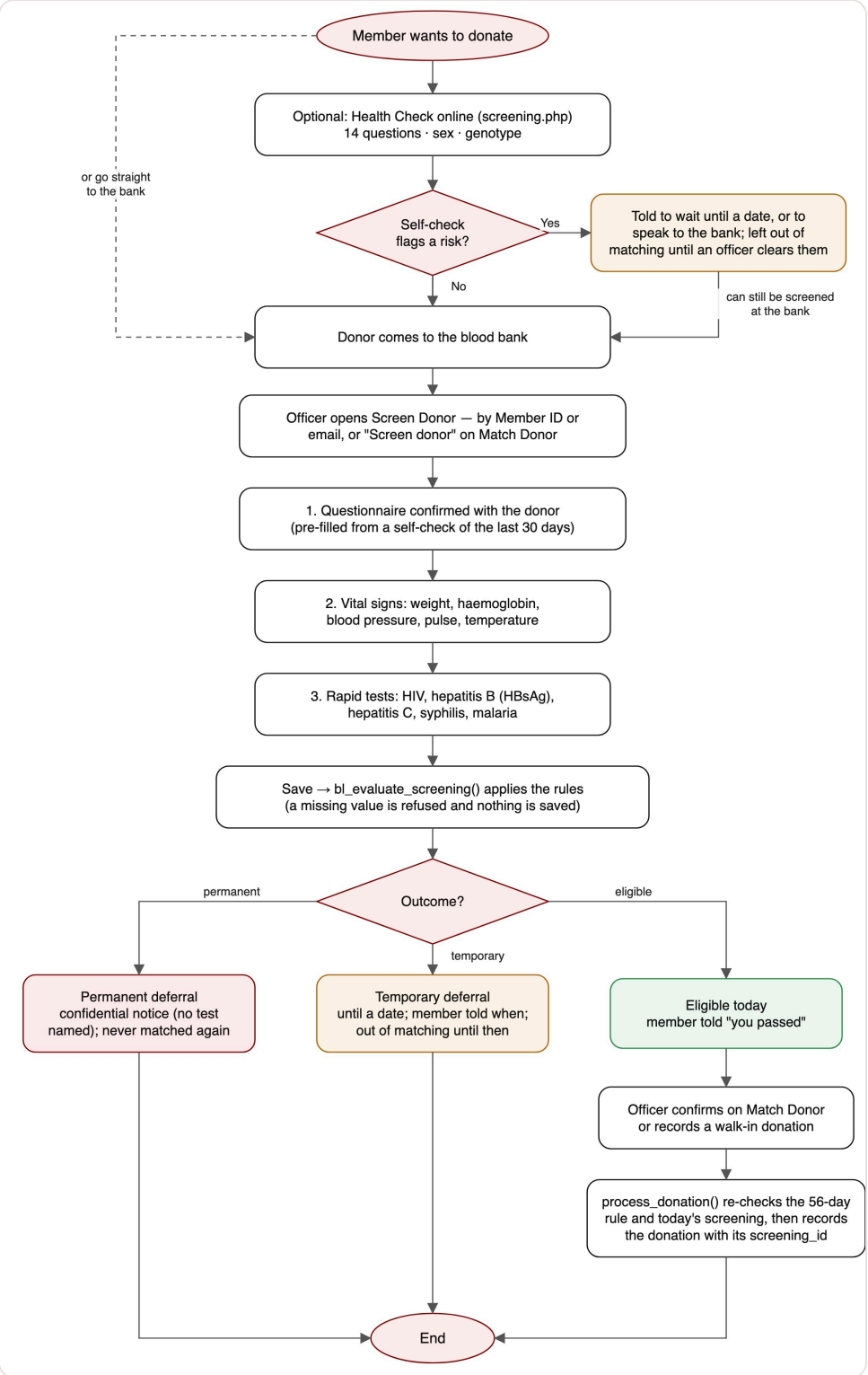
SYSTEM · PROCESS_DONATION()

### Enforces it

Match Donor shows **Confirm** only for donors who passed today; walk-ins are refused without a screening. On confirming, the 56-day rule and today's screening are re-checked and the donation is stored with its `screening_id`.

02 / The flow

# From "I want to donate" to a recorded donation




The online Health Check is optional; the officer screening at the bank is required. Deferred donors are left out of donor matching until the deferral ends.

03 / The rules

# What defers a donor, and for how long

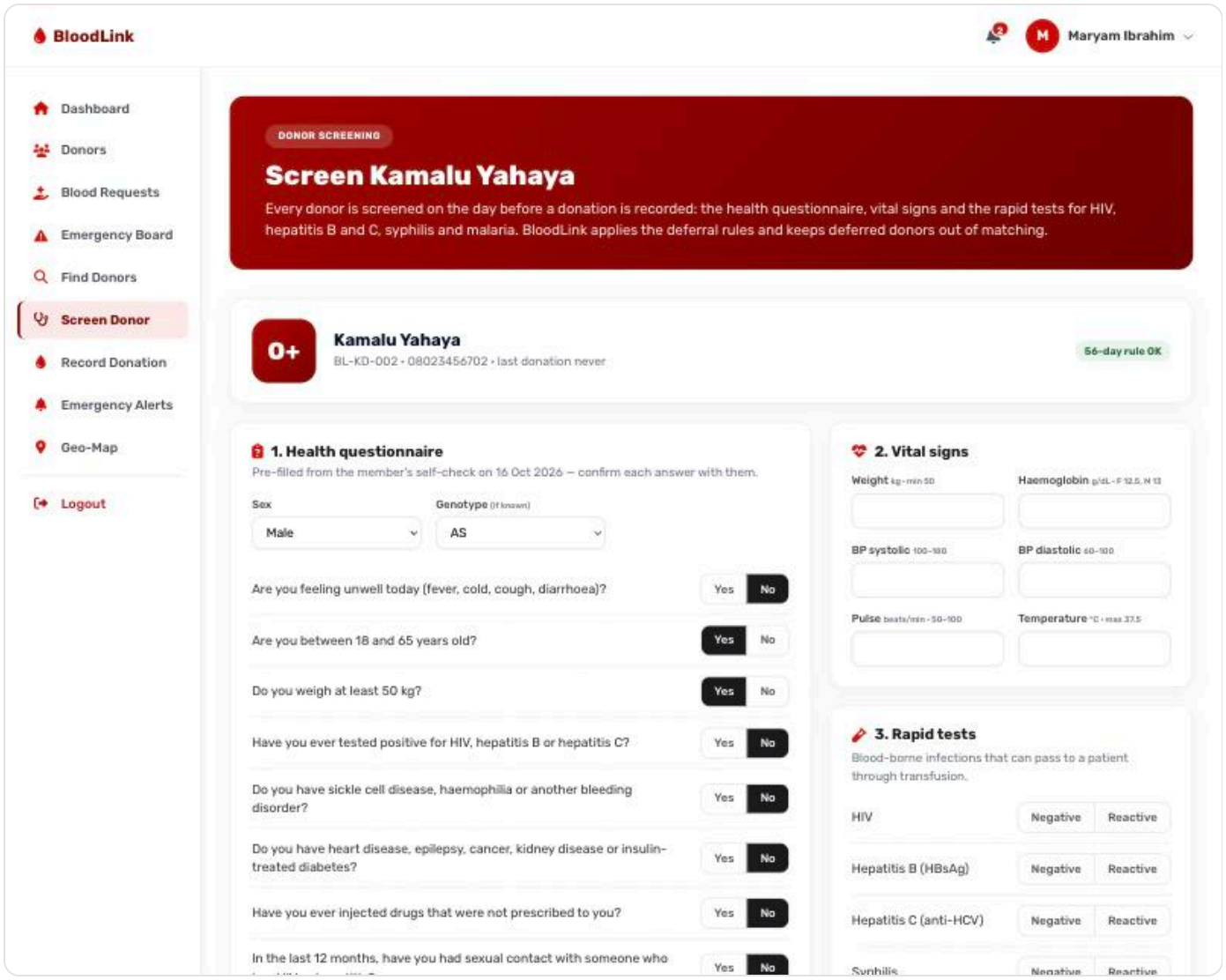
| FINDING | OUTCOME |
|---|---|
| Reactive HIV, hepatitis B (HBsAg) or hepatitis C test · ever tested positive for HIV / hepatitis · sickle cell disease or a bleeding disorder (or genotype SS / SC) · serious chronic illness (heart disease, epilepsy, cancer, kidney disease, insulin-treated diabetes) · ever injected non-prescribed drugs | **Permanent deferral** |
| Reactive syphilis test · blood transfusion, or sexual contact with someone with HIV / hepatitis, in the last 12 months · outside the 18–65 age range | Deferred 365 days |
| Pregnant, gave birth or breastfeeding in the last 6 months · tattoo, piercing or acupuncture in 6 months · major surgery in 6 months | Deferred 182 days |
| Weight under 50 kg | Deferred 90 days |
| Malaria in the last 4 weeks, or malaria parasites seen · haemoglobin below 12.5 g/dL (women) / 13.0 g/dL (men) | Deferred 28 days |
| Unwell today (fever, cold, cough, diarrhoea) · temperature above 37.5 °C | Deferred 14 days |
| Antibiotics in the last 7 days · blood pressure outside 100–180 / 60–100 · pulse outside 50–100 | Deferred 7 days |
| None of the above, every vital sign entered and all five tests negative | **Eligible** — for a donation recorded the same day |

WHICH SCREENING COUNTS

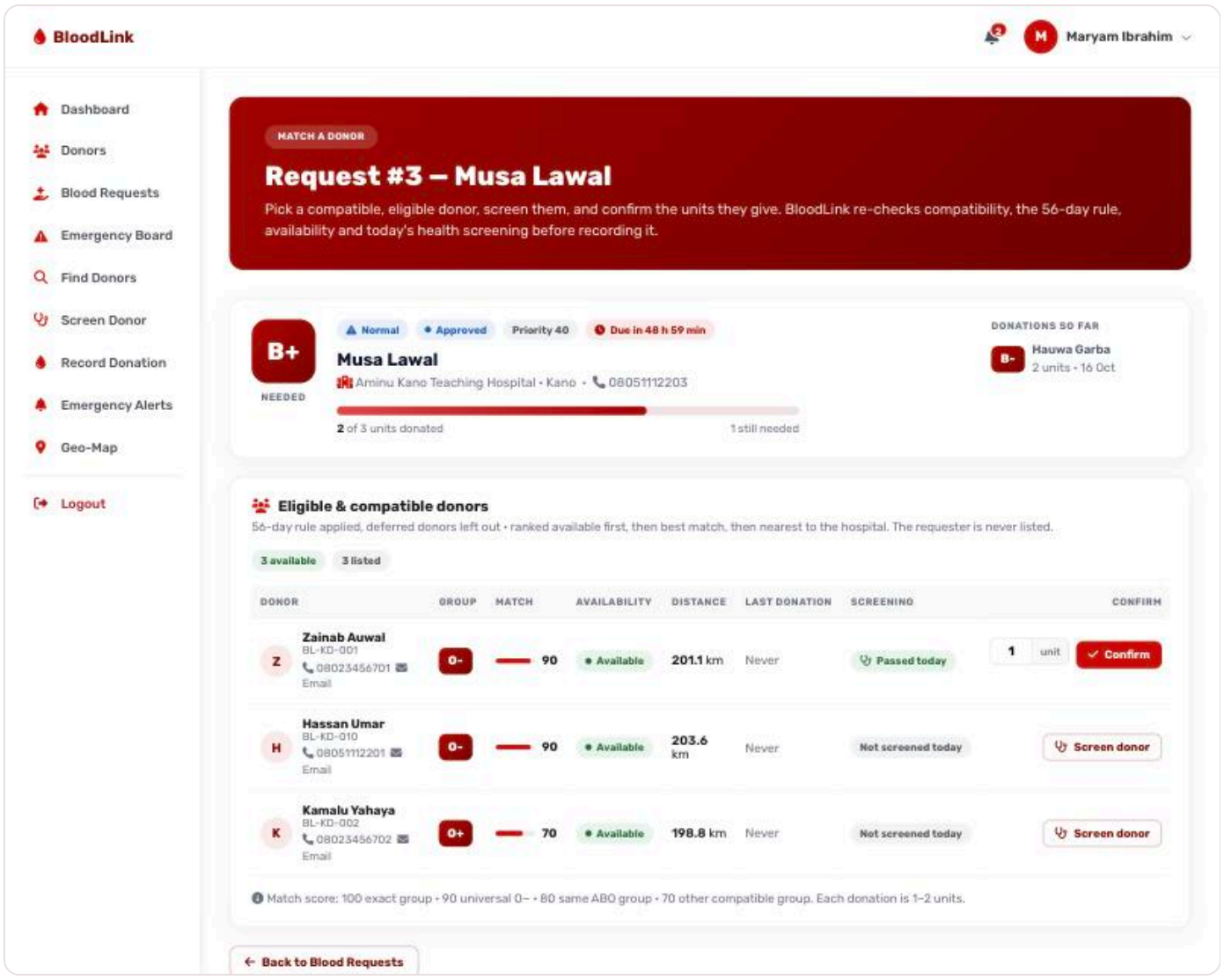
- An **officer** screening is the only one that can clear a donation, and only on the day it was done (one screening per donation).
- A member's own **Health Check** can defer them (they're left out of matching) until they retake it with clear answers, or until an officer screens them as eligible.
- A **permanent** deferral from an officer screening stays in place, even after a later clear screening or self-check.
- A **temporary** deferral ends on its date; a newer officer screening replaces an older one.

04 / What people see

# Screens and notifications




**Screen Donor** — the member card (blood group, last donation, 56-day status, any deferral), the questionnaire pre-filled from the member's Health Check, vital signs and rapid tests.




**Match Donor** — the **Screening** column shows "Passed today" (with Confirm) or "Not screened today" (with Screen donor).

| OUTCOME | THE MEMBER IS NOTIFIED | THE OFFICER SEES |
|---|---|---|
| Eligible | "You passed today's health screening — you are cleared to donate today at ..." | "... is eligible to donate today", links to Match Donor and walk-in |
| Temporary deferral (not an infection) | "You can donate again from *date*" — "Today's screening found: *reason*." | The reasons and the date |
| Temporary deferral (syphilis or malaria) | "You can donate again from *date*" — "Please speak to the team at ... — they will explain in private." | The reactive test and the date |
| Permanent deferral | "About your health screening" — "Please contact ... for a confidential discussion of your screening result." | The reasons; refer for counselling and confirmatory testing |

Notifications never name an infection test. Members see their status on **Health Check** and **Donate Blood** ("Not able to donate" or "Deferred until *date*"); officers see the reasons on Screen Donor, and deferred donors show as **Deferred** in Find Donors.

05 / Test it

# Test script

Use the seed data (QA Manual Section 09): officer `maryam@bloodlink.test` / `Officer@123`, members `…@bloodlink.test` / `Member@123`. In the seed, **Zainab** is screened on the day the seed is loaded, **Kamalu** has only done the self-check, and **Grace**'s self-check flagged malaria (deferred). On a real database nobody is screened yet, so start with O12.

## Member M13

### M13 Take the pre-donation health check

WHO Member START Sidebar → **Health Check** (or Donate Blood → **Take the health check**)

TEST DATA `yusuf@bloodlink.test`; Grace's seeded check already flags malaria.

DO

1. Choose your sex (women also see the pregnancy question) and answer every question with all-clear answers; click **Check my answers**.
2. Take it again answering **Yes** to the tattoo question.
3. Open **Donate Blood**; then retake the check with all-clear answers.

EXPECT

- ✓ "Nothing in your answers stops you donating", with what the bank will check; Donate Blood shows "Health check: Done".
- ✓ "Please wait before donating — until *date 6 months ahead*" listing the reason; you stop appearing in donor matching and Find Donors shows you as Deferred.
- ✓ Donate Blood says you're not being matched; the clear retake lifts it. A permanent deferral recorded by an officer is not lifted by a self-check ("Not able to donate").

## Hospital officer 012 · 013 · 04 · 07

### 012 Screen a donor

WHO Officer START Sidebar → **Screen Donor** (or **Screen donor** on Match Donor)

TEST DATA `BL-KD-002` (Kamalu, self-check done).

DO

1. Enter a Member ID or email and click **Start screening**.
2. Confirm the pre-filled answers, enter weight, haemoglobin, blood pressure, pulse and temperature, mark all five tests Negative; click **Save screening**.
3. Try saving with a vital sign left empty.

EXPECT

- ✓ The member card shows blood group, last donation, 56-day status and any current deferral; answers are pre-filled from their Health Check of the last 30 days.
- ✓ "... is eligible to donate today" with links to Match Donor and walk-in; the member gets "You passed today's health screening"; the screening appears under Recent screenings. From Match Donor you are taken straight back to the request.
- ✓ An empty vital sign is refused ("Enter the weight (kg)") and nothing is saved.

### 013 Screening that defers a donor

WHO Officer START `screen_donor.php`

TEST DATA `fatima@bloodlink.test` (A−), `kamalu@bloodlink.test`.

DO

1. Screen a woman with haemoglobin 11.9.
2. Screen another donor with the HIV test marked **Reactive**.
3. Try to record a walk-in donation for that donor, then screen them again with everything clear.

EXPECT

- ✓ "... is deferred until *28 days ahead*" listing "Haemoglobin 11.9 g/dL is below 12.5"; the donor is told when they can donate again.
- ✓ "... is permanently deferred" — refer for confidential counselling. The member's notification says only "About your health screening ... confidential discussion" and never names the test.
- ✓ The walk-in is refused ("deferred from donating"). A later clear screening does not lift a permanent deferral; the donor no longer appears in Match Donor and shows as Deferred in Find Donors.

## 04 Confirm a donation

WHO Officer START `match_donor.php`

TEST DATA donor `Zainab Auwal` (O−) — the seed already screened her today.

DO

1. On request #2 (AB+, 1 unit), leave the units box at 1 and click **Confirm** next to an O− donor.
2. Accept the prompt "Record 1 unit(s) from … for this request?".

EXPECT

- ✓ "Donation recorded (1 unit(s) of O-). Request is now Fulfilled." — recorded under the **donor's** group.
- ✓ The donor gets "Thank you — your donation was recorded"; the requester gets "Your blood request is Fulfilled".
- ✓ The request leaves the queue; opening it again shows "no further donations can be recorded". Bank stock is unchanged (directed blood goes to the patient).

## 07 Record a walk-in donation

WHO Officer START Sidebar → **Record Donation**

TEST DATA `kamalu@bloodlink.test` (O+).

DO

1. Enter a member's email or Member ID and 1 unit; click **Record Donation** and confirm.
2. Click **Screen this donor now**, complete an all-clear screening, then record the donation again.
3. Try the same member again; try an unknown email.

EXPECT

- ✓ The first try is refused: "This donor has not passed a health screening today" with a **Screen this donor now** link.
- ✓ After screening: "Walk-in donation recorded: 1 unit(s) of … added to your bank's stock." and it appears under Recent walk-in donations.
- ✓ Your bank's stock for that group rises (O1); a low-stock alert clears once stock is back above threshold.
- ✓ The member is thanked and sees it in My Donations. A repeat is refused by the 56-day rule; unknown members show "No member found".

IF SOMETHING LOOKS WRONG

- **No Confirm button on Match Donor** — the donor hasn't passed a screening today: click **Screen donor**.
- **A member is missing from Match Donor** — they're deferred (or inside the 56 days, or unavailable): open them on **Screen Donor** to see why.
- **"has not passed a health screening today"** on a seeded donor — the seed's screening was loaded on an earlier day; screen them again.
- **"Enter the …" / "Answer every question"** — the screening is incomplete and nothing was saved.

06 / Where it's stored

# The donor_screenings table

| COLUMN(S) | HOLDS |
|---|---|
| member_id | The donor (deleting the member deletes their screenings) |
| source | `self` (Health Check) or `officer` (Screen Donor) |
| officer_id, blood_bank_id | Who screened and where (empty for a self-check) |
| answers | The questionnaire as JSON (question key → yes / no) |
| sex, genotype | Set the haemoglobin limit and the pregnancy question; SS / SC defer permanently |
| weight_kg … temperature_c | Weight, haemoglobin, blood pressure (two columns), pulse, temperature |
| hiv_result … malaria_result | negative / positive / not_tested for the five rapid tests |
| outcome, deferral_reasons, deferred_until | The decision, why, and until when |
| donations.screening_id | Links each donation to the screening that cleared it |

Existing databases get the table with `db/upgrade_2026_10_screening.sql` (safe to run more than once). The rapid tests are screening tests; confirmatory laboratory testing of the collected blood is outside BloodLink.